

# **VTU Belagavi**

## **REGULAR EXAMINATION ? 2023**

Power Systems-I ? Regular Examination, 2023

Subject Code: EE302 Semester: 4 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Transmission Line Parameters)**

1. Derive the expression related to Transmission Line Parameters and explain each term. [8 Marks]
2. Define Transmission Line Parameters. Explain its significance in Power Systems-I with a suitable example. [12 Marks]
3. Solve a problem related to Transmission Line Parameters. Show all the steps clearly. [8 Marks]

### **SECTION B (Unit 2: Transmission Line Performance)**

4. Write short notes on Transmission Line Performance. [12 Marks]
5. Explain Transmission Line Performance with a neat diagram/sketch and a suitable example. [12 Marks]
6. Describe the working principle of Transmission Line Performance with necessary equations. [12 Marks]

### **SECTION C (Unit 3: Underground Cables)**

7. Derive the expression related to Underground Cables and explain each term. [12 Marks]
8. List the advantages and limitations of Underground Cables. Explain any two in detail. [12 Marks]
9. Discuss the engineering applications of Underground Cables. [12 Marks]

### **SECTION D (Unit 4: Distribution Systems)**

10. How is Distribution Systems implemented? Give an algorithm / procedure and explain. [12 Marks]
11. List the advantages and limitations of Distribution Systems. Explain any two in detail. [8 Marks]
12. Describe the working principle of Distribution Systems with necessary equations. [12 Marks]

### **SECTION E (Unit 5: Power System Protection)**

13. Write short notes on Power System Protection. [12 Marks]
14. Solve a problem related to Power System Protection. Show all the steps clearly. [8 Marks]

**Marks]**

15. Compare and contrast the key concepts involved in Power System Protection. [12 Marks]

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